Investment Banking with Finance Capstone Project

Project Title

Investment Recommendation System

Project Overview

The Investment Recommendation System is a finance-based analytical application designed to help users analyze investment options and generate recommendations based on financial data, risk levels, and market trends. The system evaluates stocks, mutual funds, or investment portfolios and provides insights through dashboards and reports.

This project gives students practical exposure to financial analytics, investment decision-making, and data-driven recommendation systems used in banking and finance industries.

Project Objective

Develop a finance analytics application that can:

- Analyze investment data
- Compare financial assets
- Evaluate risk and return
- Generate investment recommendations
- Visualize market trends and portfolio insights

Industry Relevance

This project is inspired by systems used in:

- Investment banking firms
- Wealth management companies
- Financial advisory platforms
- Stock market analytics tools
- FinTech applications

Students will understand how financial analytics supports investment decisions.

Project Modules

Module 1: Financial Data Collection

Collect and manage:

- Stock prices
- Mutual fund data
- Market trends
- Historical financial records

Learning Outcomes

- Financial data handling
- API/data integration
- Data preprocessing

Module 2: Investment Analysis

Analyze:

- Risk levels
- Return percentages
- Price trends
- Portfolio performance
- Market movement

Learning Outcomes

- Financial analysis
- Risk assessment
- Data interpretation

Module 3: Recommendation System

Generate recommendations such as:

- Buy
- Hold
- Sell

Based on:

- Risk score
- Market trends
- Historical performance

Learning Outcomes

- Recommendation logic
- Decision-support systems
- Financial analytics

Module 4: Dashboard & Visualization

Display:

- Investment performance
- Stock comparison
- Portfolio analysis
- Risk indicators
- Trend charts

Learning Outcomes

- Dashboard development
- Data visualization
- Financial reporting

Module 5: Report Generation

Generate reports including:

- Investment summaries
- Portfolio insights
- Risk analysis reports

Learning Outcomes

- Report automation
- Business presentation
- Financial documentation

Recommended Tech Stack

Programming

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Plotly
- Scikit-learn (Optional)

Dashboard

- Streamlit
- Power BI

Database

- MySQL / Firebase

Deployment

- Streamlit Cloud
- Render

Project Workflow

1. Financial Data Collection
2. Data Cleaning & Analysis
3. Risk & Return Evaluation
4. Recommendation Generation
5. Dashboard Visualization
6. Report Generation

Dataset Suggestions

Students may use:

- Yahoo Finance data
- Kaggle stock market datasets
- NSE/BSE historical datasets

Deliverables

Students must submit:

- Source Code
- Project Report
- PPT Presentation
- GitHub Repository
- Dashboard Screenshots
- Deployment Link
- Demo Video

Scope Limitations

Students should avoid:

- Real-time trading systems
- Advanced quantitative finance models
- Live brokerage integration
- High-frequency trading concepts

Focus should remain on building a clean and functional investment analytics system.

25-Day Project Plan

Days	Tasks
Day 1–3	Requirement Analysis and Dataset Collection
Day 4–7	Financial Data Processing
Day 8–12	Investment & Risk Analysis
Day 13–17	Recommendation System Development
Day 18–21	Dashboard Development
Day 22–23	Report Generation and Testing
Day 24	Bug Fixing and Optimization
Day 25	Final Deployment and Presentation

Evaluation Criteria

Criteria	Weightage
Functionality	30%
Financial Analysis	25%
Recommendation Logic	20%
Dashboard/UI	10%
Deployment	10%
Documentation	5%

Final Outcome

By completing this project, students will learn:

- Financial data analysis
- Investment risk evaluation
- Recommendation system development
- Dashboard creation
- Data visualization
- End-to-end finance analytics workflow

The final system will function as an Investment Recommendation platform capable of analyzing financial data, evaluating investment opportunities, and generating investment insights and recommendations.